

Re-analysis Summary Report for Peer-evaluation

Paper ID: **Bruner_ExpEco_2017_amYY**

Re-analyst's ID: **7nm5s**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: <https://forms.gle/EAREaC6JjN2PWFfR6>

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
the likelihood ... [of decision error] should decrease with the degree of risk aversion (p. 269.)

The corresponding article

Here you can find the original article: https://osf.io/2efb6/?view_only=c874fcf097644e82a320679d4641163a
and the original data: https://osf.io/8xq95/?view_only=f00ab18c5cf94087994e3bb0740ddad1

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:

Participants performed three tasks in this study - a probability variation (PV), risk variation (RV), and lottery variation (LV) task. Each task consisted of 10 items making participants choose between two options. The claim "the likelihood ... [of decision error] should decrease with the degree of risk aversion" of the original paper has been transformed into the operational hypothesis "More frequent choices of safe options in both the risk and probability variation tasks are associated with fewer violations of stochastic dominance in the lottery variation task." and the statistical hypothesis "The average number of safe choices in the RV and PV tasks are negatively correlated with the number of violations of stochastic dominance in the LV task." Thus, risk aversion was operationalized by the number of choosing the safe over the risky option in the RV and PV tasks. Therefore, the number of times choosing the safe option was summed up for each task before calculating the arithmetic mean of safe choices across both tasks. Decision errors were operationalized as violations of stochastic dominance in the LV task. As all options in that task have the same expected value, dominance is defined (as in the original paper) as "the asset with the lower potential return but greater chance of success", that is, by a higher chance of receiving an outcome even if this outcome has a lower magnitude than the alternative. Thus, each response can be categorized as a violation of stochastic dominance (i.e., a decision error) or no violation (except for item 5, in which both options are identical, which is why this item

is excluded from the score of decision errors). The number of violations was summed up over all nine eligible items. As both variables (average number of safe choice in the PV and RV tasks, number of decision errors) are count data without normal distribution, Pearson correlation would be suboptimal. Furthermore, there are many ties in the data, which Spearman correlation cannot handle well. Therefore, Kendall's tau was computed as the measure of association between risk aversion and decision errors. The level of statistical significance was set to $\alpha=0.05$. This test resulted in a statistically significant negative correlation of Kendall's $\tau=-.36$, $CI=[-.48, -.22]$, $p<.001$, signifying higher risk aversion being associated with fewer decision errors.

The analyst's conclusion: Higher risk aversion (i.e., more frequently favoring safe over risky options) was associated with fewer decision errors (i.e., fewer violations of stochastic dominance in a lottery task, in which both options were risky).

The analyst's categorization of the result:

The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:

You should not use rank-order tests in your analysis.

Here, the analyst reported the following steps and results of the analysis:

Participants performed three tasks in this study - a probability variation (PV), risk variation (RV), and lottery variation (LV) task. Each task consisted of 10 items making participants choose between two options. The claim "the likelihood ... [of decision error] should decrease with the degree of risk aversion" of the original paper has been transformed into the operational hypothesis "Risk averse participants (with more frequent choices of safe options in both the risk and probability variation tasks) fewer violations of stochastic dominance in the lottery variation task than risk seeking participants (with less frequent choices of safe options in both the risk and probability variation tasks)." and the statistical hypothesis "Risk averse participants (with an average number of safe choices in RV and PV tasks greater than five) have statistically significantly fewer violations of stochastic dominance in the LV task than risk seeking participants (with an average number of safe choices in RV and PV tasks smaller or equal to five)." Thus, risk aversion was operationalized by the number of choosing the safe over the risky option in the RV and PV tasks. Therefore, the number of times choosing the safe option was summed up for each task before calculating the arithmetic mean of safe choices across both tasks. This mean was used to divide the sample in risk averse participants with more than five safe choices on average in each task and risk seeking participants with five or fewer safe choices. Decision errors were operationalized as violations of stochastic dominance in the LV task. As all options in that task have the same expected value, dominance is defined (as in the original paper) as "the asset with the lower potential return but greater chance of success", that is, by a higher chance of receiving an outcome even if this outcome has a lower magnitude than the alternative. Thus, each response can be categorized as a violation of stochastic dominance (i.e., a decision error) or no violation (except for item 5, in which both options are identical, which is why this item is excluded from the score of decision errors). The number of violations was summed up over all nine eligible items. To satisfy the additional constraint of not using rank order tests, a Pearson correlation between the average number of safe choices in PV and RV tasks and the number of decision errors in the LV task was calculated. The level of statistical significance was set to $\alpha=0.05$. This test resulted in a statistically significant negative correlation of $r=-.46$, $CI=[-.60, -.29]$, $p<.001$, signifying higher risk aversion being associated with fewer decision errors.

Analysis language/software/code (optional to check):

Task 1: R

Task 2: R

Analysis codes: <https://osf.io/jae5q/>